

6.18 Solution: (a) Given a point $\mathbf{x}' = (0, 0, z')$ on the negative z -axis, $z' < 0$, $d\mathbf{l}' = dz'\hat{k}$, and $\mathbf{x} - \mathbf{x}' = (x, y, z - z')$, then

$$d\mathbf{l}' \times (\mathbf{x} - \mathbf{x}') = (-y\hat{i} + x\hat{j})dz'.$$

The vector potential can be calculated as

$$\begin{aligned}\mathbf{A}(\mathbf{x}) &= \frac{g}{4\pi} \int_L \frac{d\mathbf{l}' \times (\mathbf{x} - \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|^3} \\ &= \frac{g}{4\pi} (-y\hat{i} + x\hat{j}) \int_{-\infty}^0 \frac{dz'}{[x^2 + y^2 + (z - z')^2]^{3/2}} \\ &= \frac{g}{4\pi} (-y\hat{i} + x\hat{j}) \frac{1}{x^2 + y^2} \left. \frac{z' - z}{[x^2 + y^2 + (z - z')^2]^{1/2}} \right|_{z'=-\infty}^0 \\ &= \frac{g}{4\pi} (-y\hat{i} + x\hat{j}) \frac{1}{x^2 + y^2} \left(1 - \frac{z}{[x^2 + y^2 + z^2]^{1/2}} \right).\end{aligned}$$

In spherical coordinates, the vector potential is

$$\begin{aligned}\mathbf{A}(\mathbf{x}) &= \frac{g}{4\pi} r \sin \theta (-\sin \phi \hat{i} + \cos \phi \hat{j}) \frac{1}{r^2 \sin^2 \theta} \left(1 - \frac{r \cos \theta}{r} \right) \\ &= \frac{g}{4\pi r} \frac{1 - \cos \theta}{\sin \theta} (-\sin \phi \hat{i} + \cos \phi \hat{j}) \\ &= \frac{g}{4\pi r} \frac{1 - \cos \theta}{\sin \theta} \hat{\phi} \\ &= \frac{g}{4\pi r} \tan \left(\frac{\theta}{2} \right) \hat{\phi},\end{aligned}$$

where we have used $\hat{\phi} = -\sin \phi \hat{i} + \cos \phi \hat{j}$.

(b) The magnetic field is

$$\mathbf{B} = \nabla \times \mathbf{A} = \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\phi) \hat{r} - \frac{1}{r} \frac{\partial}{\partial r} (r A_\phi) \hat{\theta} = \frac{g}{4\pi r^2} \hat{r}.$$

This resembles the electric field generated by a point charge. Note that the vector potential is singular at $\theta = \pi$, or on the negative z -axis, as our integration path is along it. Therefore, at $\theta = \pi$, the magnetic field is not well-defined.

(c) For $0 < \theta < \pi/2$, on the disk, for a point at (ρ, ϕ, z) , the magnetic flux around this point is

$$\mathbf{B} \cdot \mathbf{n} da = \frac{g}{4\pi(\rho^2 + z^2)} \cos \alpha \cdot \rho d\rho d\phi = \frac{gz}{4\pi(\rho^2 + z^2)^{3/2}} \rho d\rho d\phi,$$

where α is the angle between the direction from the origin to this point and the z -axis. Therefore, the total magnetic flux through the disk is

$$\begin{aligned}\Phi_{>} &= \oint \mathbf{B} \cdot \mathbf{n} da = \frac{g}{4\pi} \int_0^{2\pi} d\phi \int_0^{R \sin \theta} d\rho \frac{z\rho}{(\rho^2 + z^2)^{3/2}} \\ &= -\frac{g}{2} \left. \frac{z}{(\rho^2 + z^2)^{1/2}} \right|_{\rho=0}^{R \sin \theta} = \frac{g}{2} \left(1 - \frac{z}{\sqrt{z^2 + R^2 \sin^2 \theta}} \right)\end{aligned}$$

$$= \frac{g}{2} \left(1 - \frac{z}{R}\right).$$

For $z < 0$, the downward flux is given by the same result, with z replaced by $-z$. Then, the upward flux is the negative of it,

$$\Phi_{<} = -\frac{g}{2} \left(1 + \frac{z}{R}\right).$$

(d) Since the vector potential has only ϕ -component, the line integral can be easily evaluated as

$$\oint \mathbf{A} \cdot d\mathbf{l} = \frac{g}{4\pi R} \int_0^{2\pi} \frac{1 - \cos \theta}{\sin \theta} R \sin \theta d\phi = \frac{g}{2} (1 - \cos \theta) = \frac{g}{2} \left(1 - \frac{z}{R}\right).$$

Comparing with the results from part (c), we can see that this agrees with the magnetic flux for $0 < \theta < \pi/2$, but differs a constant of g for $\pi/2 < \theta < \pi$. The difference is due to the loop around the negative z -axis also encloses the string, which is singular in the contribution to the magnetic field and the magnetic flux.